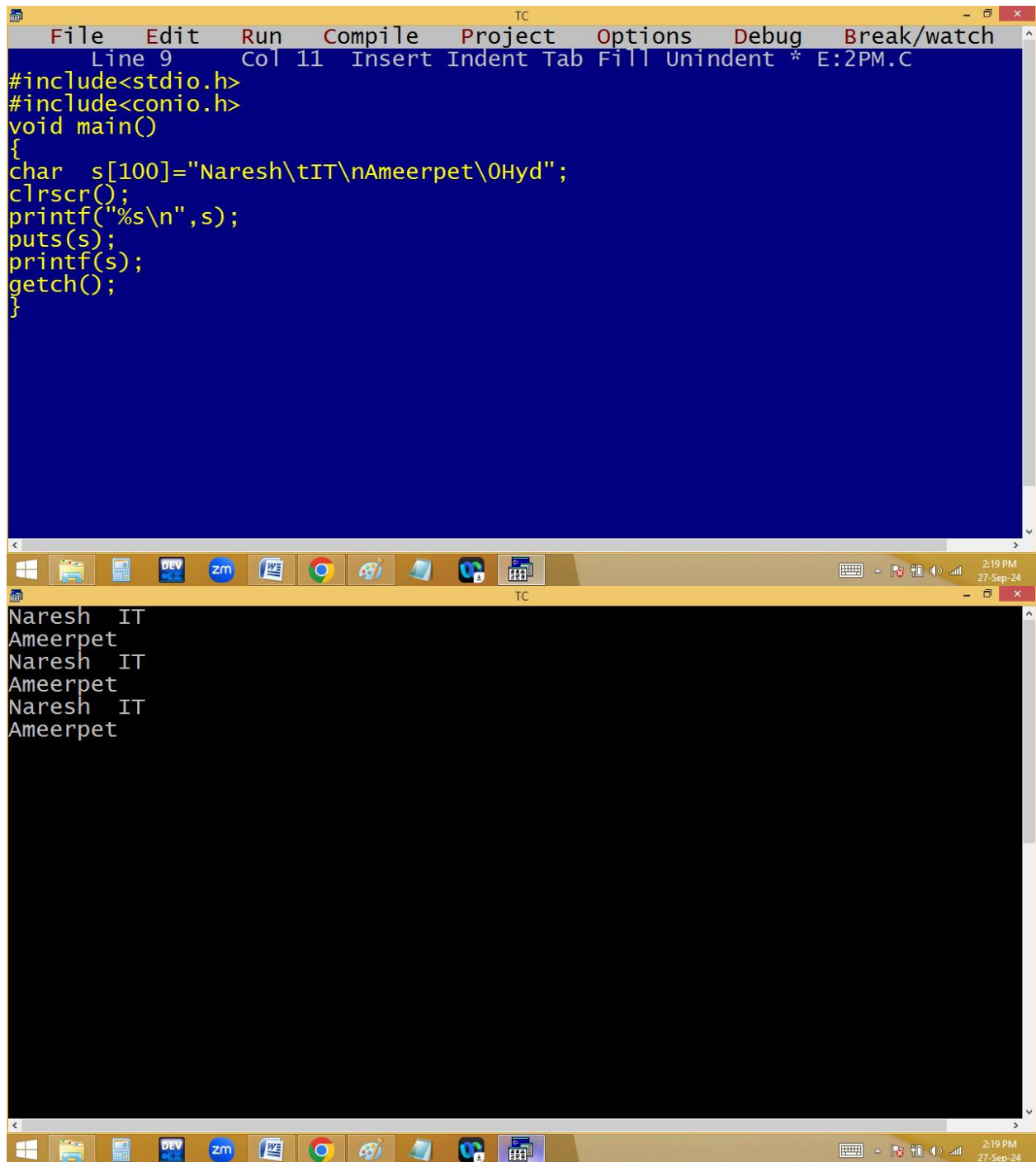


Direct initialization of string:



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a program that demonstrates direct initialization of a string. The code includes `<stdio.h>` and `<conio.h>`, defines a `main` function, and declares a character array `s` of size 100. The string `"Naresh\tIT\nAmeerpet\0Hyd"` is assigned to `s` using direct initialization. The program then calls `clrscr()`, `printf("%s\n", s);`, `puts(s);`, `printf(s);`, and `getch();`. The bottom window shows the output of the program, which displays the string `Naresh IT` on the first line and `Ameerpet` on the second line, followed by another instance of `Naresh IT` and `Ameerpet` on the next two lines. The Windows taskbar at the bottom indicates the time is 2:19 PM on 27-Sep-24.

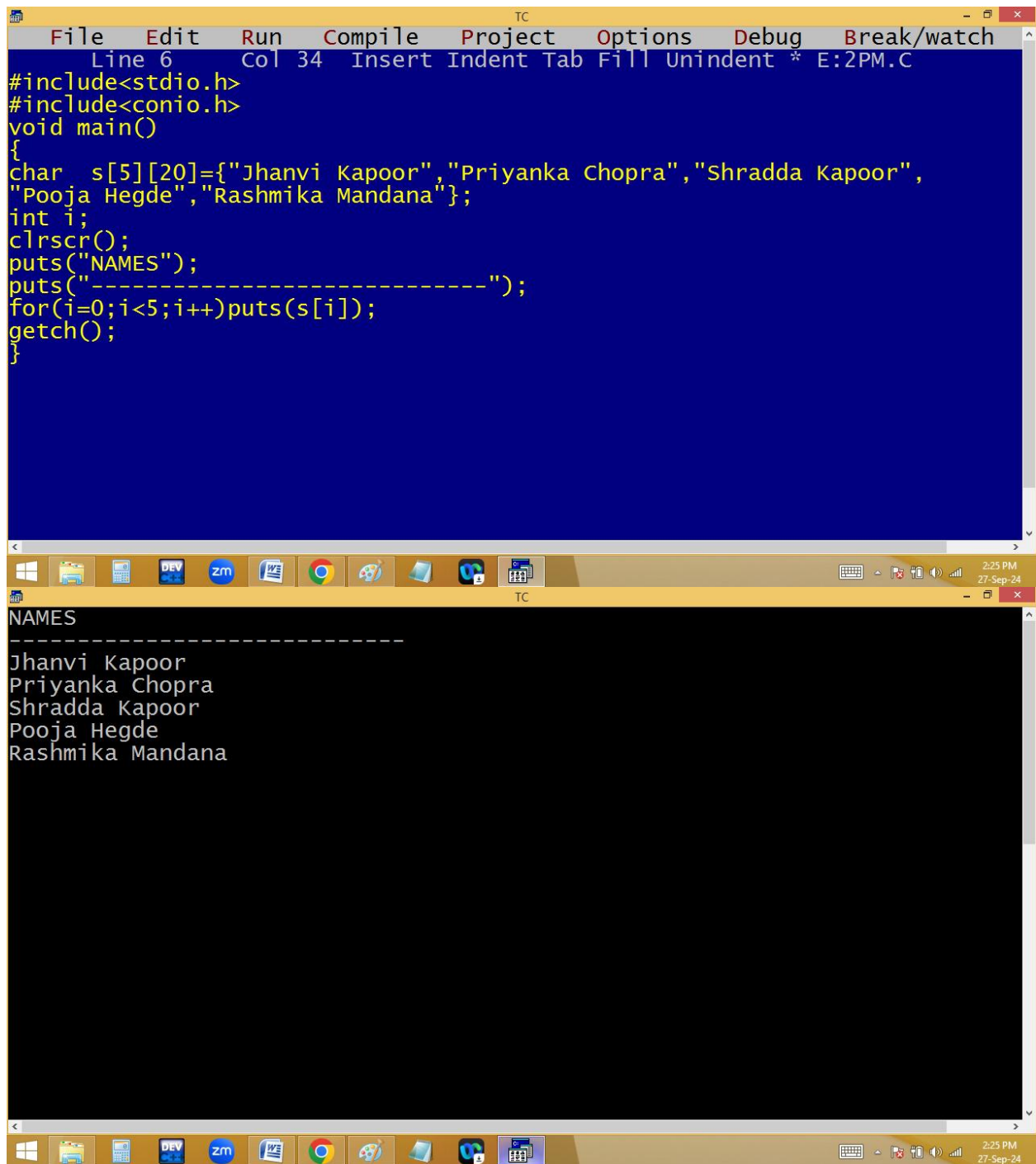
```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]="Naresh\tIT\nAmeerpet\0Hyd";
clrscr();
printf("%s\n",s);
puts(s);
printf(s);
getch();
}
```

Naresh IT
Ameerpet
Naresh IT
Ameerpet
Naresh IT
Ameerpet

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s1[2]="N",s2[]="N",s3[2]={'N'},s4[]={'N','\0'},s5[]={'N'};
clrscr();
puts(s1);
puts(s2);
puts(s3);
puts(s4);
puts(s5);_
getch();
}
```

N
N
N
N
N
NC∞ ↔☺☹

Storing of multiple strings:



The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays the source code of a C program. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, a character array `s` of size 5x20 is initialized with the names "Jhanvi Kapoor", "Priyanka Chopra", "Shradda Kapoor", "Pooja Hegde", and "Rashmika Mandana". The program then clears the screen with `clrscr()`, prints the word "NAMES", prints a separator line of dashes, and uses a `for` loop to print each name from the array. Finally, it calls `getch()` to pause the execution.

```
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 34 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[5][20]={"Jhanvi Kapoor","Priyanka Chopra","Shradda Kapoor",
"Pooja Hegde","Rashmika Mandana"};
int i;
clrscr();
puts("NAMES");
puts("-----");
for(i=0;i<5;i++)puts(s[i]);
getch();
}
```

The bottom window shows the output of the program. It displays the word "NAMES", followed by a line of dashes, and then the five names listed vertically: "Jhanvi Kapoor", "Priyanka Chopra", "Shradda Kapoor", "Pooja Hegde", and "Rashmika Mandana".

```
NAMES
-----
Jhanvi Kapoor
Priyanka Chopra
Shradda Kapoor
Pooja Hegde
Rashmika Mandana
```

The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays the source code for a C program. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, a character array `s` of size `5x20` is initialized with five names: "Jhanvi Kapoor", "Priyanka Chopra", "Shradda Kapoor", "Pooja Hegde", and "Rashmika Mandana". The program then prints the word "NAMES", a separator line of dashes, and each character of the array in a single column using a `printf` loop. A `getch` call is used to pause the execution before the window closes. The bottom window shows the output of the program, which matches the code's output. The Windows taskbar at the bottom includes icons for various applications and the system clock showing 2:27 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 36 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[5][20]={"Jhanvi Kapoor","Priyanka Chopra","Shradda Kapoor",
"Pooja Hegde","Rashmika Mandana"};
int i;
clrscr();
puts("NAMES");
puts("-----");
for(i=0;i<5;i++)printf("%c",s[i][i]);
getch();
}
```

NAMES

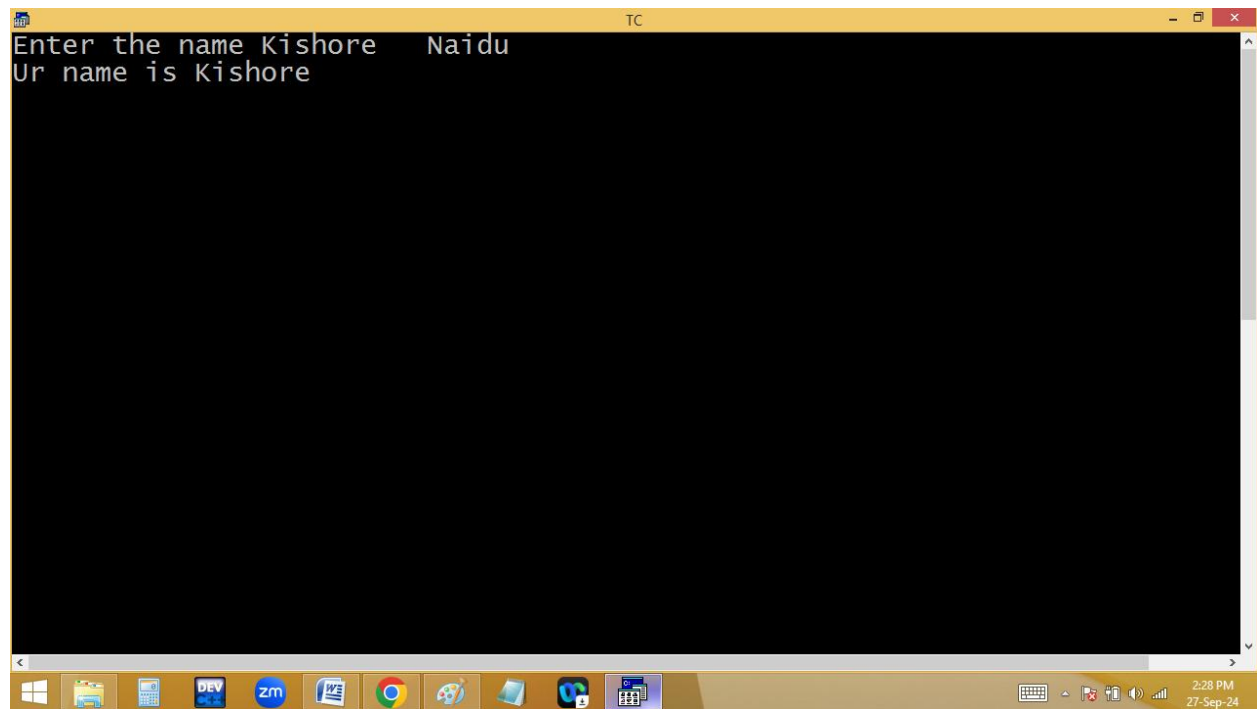
Jrrjm

Reading and printing string:

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that prompts the user for a name, reads it using `scanf`, and prints it using `printf`. The bottom screenshot shows the same IDE with the program executed, displaying the user input 'Kishore' and the output 'Ur name is Kishore_'. The Windows taskbar at the bottom of both screenshots shows various application icons and the system clock indicating 2:28 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the name "); scanf("%s",s);
printf("Ur name is %s",s);
getch();
}
```

Enter the name Kishore
Ur name is Kishore_



```
TC
Enter the name Kishore   Naidu
Ur name is Kishore
```

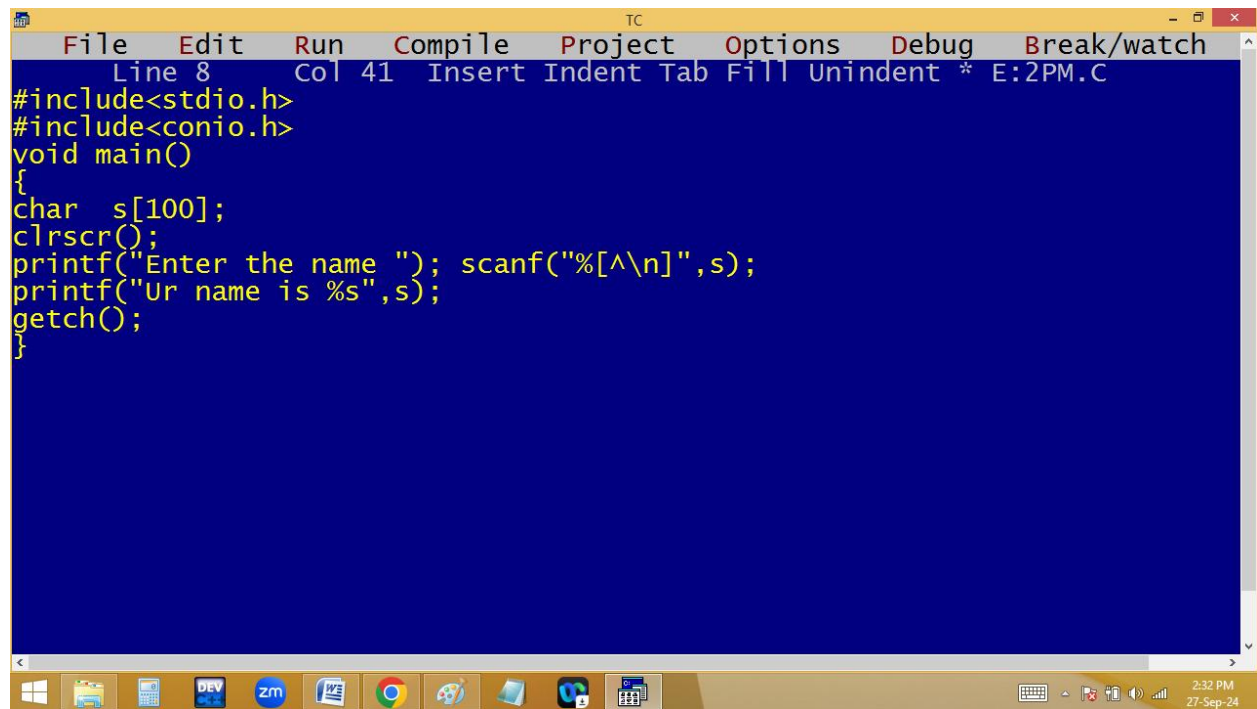
The screenshot shows a Windows desktop with a taskbar at the bottom. The taskbar contains icons for Windows, File Explorer, a calendar, DEV C++, Zoom, Word, Chrome, Paint, a folder, and a game. The system tray on the right shows the time as 2:28 PM on 27-Sep-24. The TC window is a standard application window with a yellow title bar and a black text area. The text in the window is white and shows the execution of a C program that takes 'Kishore' as input and prints 'Ur name is Kishore'.

Magic characters / scan set characters:

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that prompts the user for a name, reads it using `scanf`, and prints it back using `printf`. The bottom screenshot shows the same IDE with the program executed, displaying the user input 'Virat Kohli' and the program's output 'Ur name is Virat Kohli_'. The Windows taskbar at the bottom of both screenshots shows various application icons and the system clock indicating 2:30 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 41 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the name "); scanf("%[^\n]",s);
printf("Ur name is %s",s);
getch();
}
```

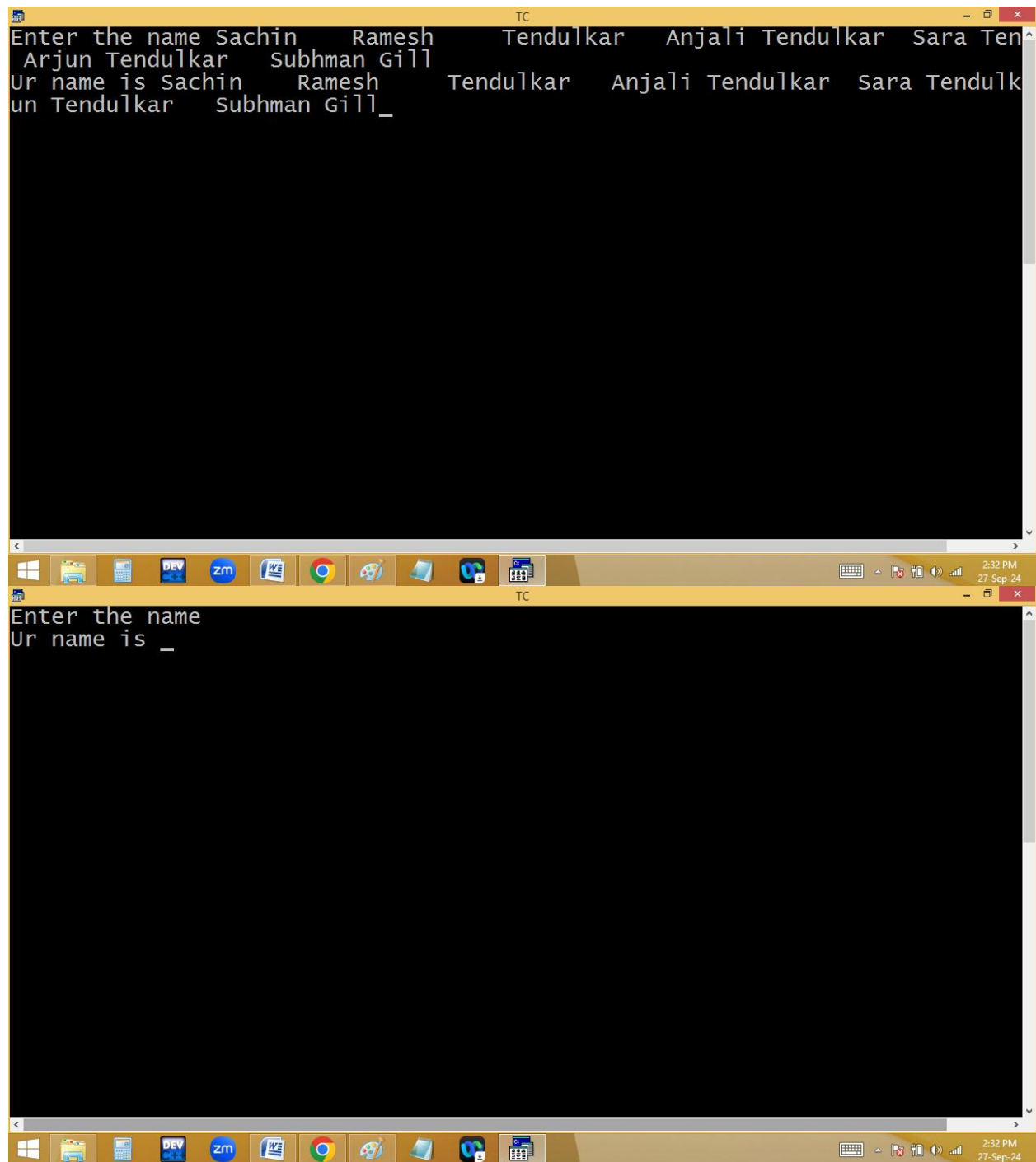
Enter the name Virat Kohli
Ur name is Virat Kohli_

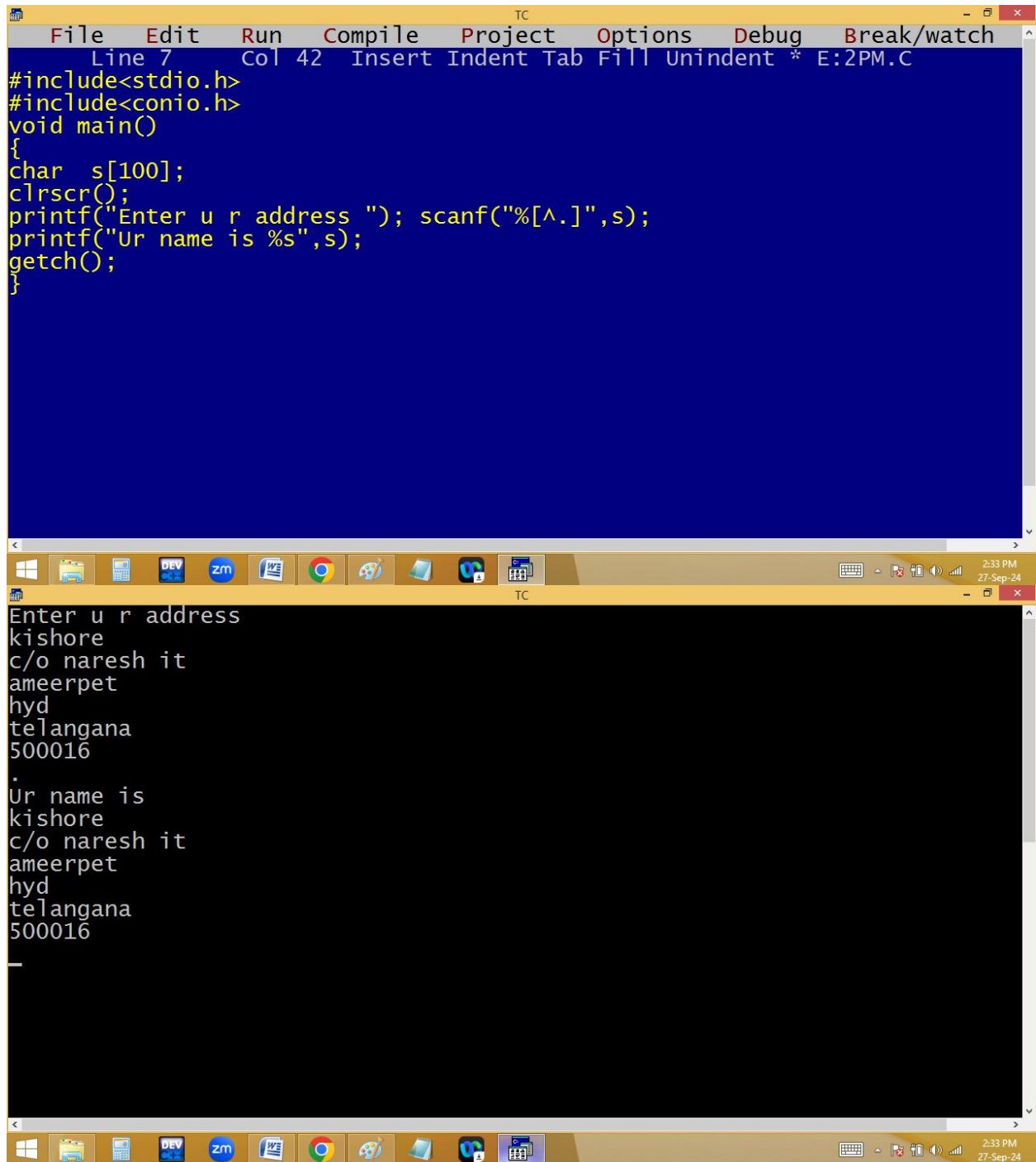


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 8", "Col 41", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the name "); scanf("%[^\n]",s);
printf("Ur name is %s",s);
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Task View, and several application icons including DEV, zm, and Chrome. The system clock in the bottom right corner shows "2:32 PM" and "27-Sep-24".





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code of a C program. The code includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside `main`, it declares a character array `s` of size 100, clears the screen with `clrscr()`, prompts the user to enter an address, reads the input using `scanf`, prints the entered address, prompts for a name, reads the name using `scanf`, and prints the entered name. The bottom window shows the execution output of the program. It displays the prompts and the user's input for both the address and the name, which are then printed back to the screen.

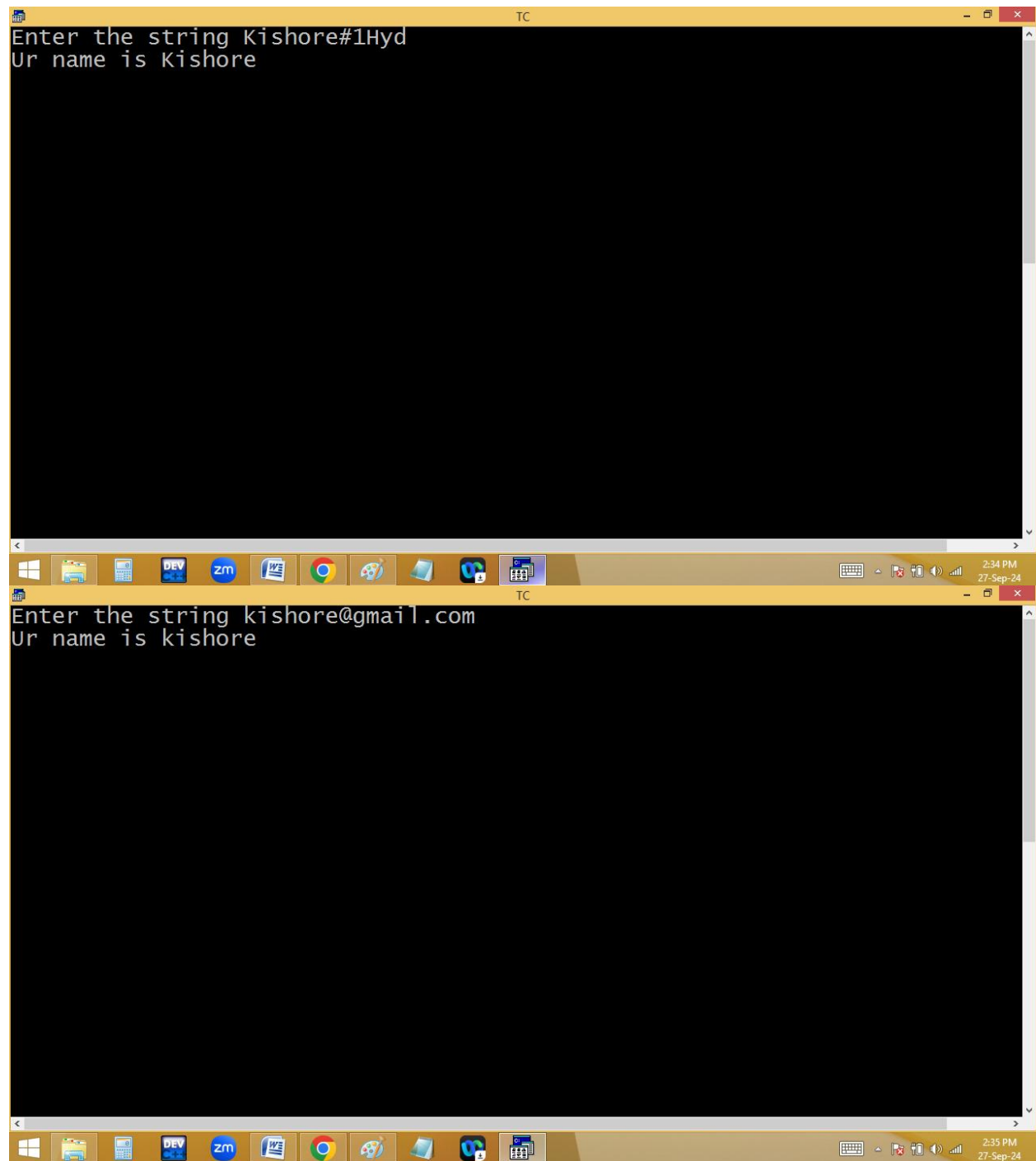
```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 42 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter u r address "); scanf("%[^\n]",s);
printf("Ur name is %s",s);
getch();
}
```

Enter u r address
kishore
c/o naresh it
ameerpet
hyd
telangana
500016
.
Ur name is
kishore
c/o naresh it
ameerpet
hyd
telangana
500016
_

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that declares a character array `s` of size 100, clears the screen, prompts the user to enter a string, reads the input using `scanf`, prints the input, and waits for a key press using `getch`. The bottom screenshot shows the same IDE with the program executed. The output window displays the prompt "Enter the string" followed by the user input "Us\$Dollers", and the program's response "Ur name is Us_". The Windows taskbar at the bottom of both screenshots shows the time as 2:34 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 43 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); scanf("%[^@#$_]",s);
printf("Ur name is %s",s);
getch();
}
```

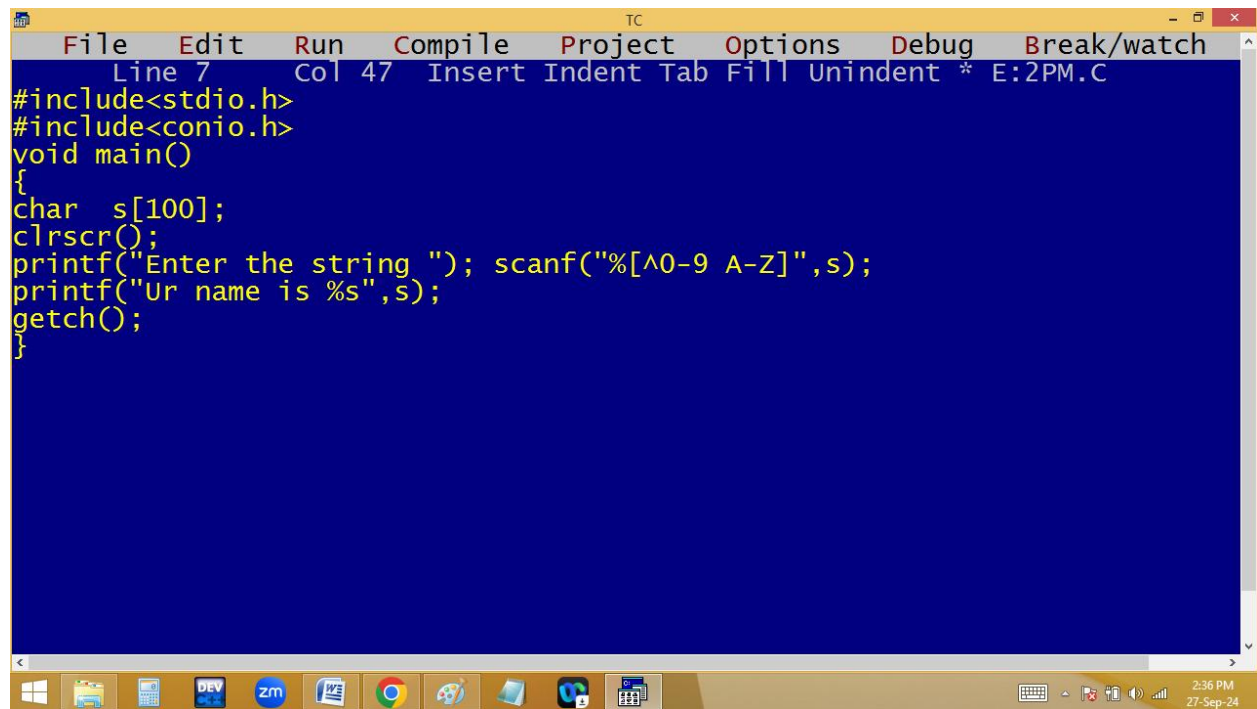
Enter the string Us\$Dollers
Ur name is Us_



The image shows two windows of the Turbo C++ (TC) IDE. The top window is the source code editor for a file named 'E:2PM.C'. It contains the following C code:

```
Line 7 Col 43 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); scanf("%[^0-9]",s);
printf("Ur name is %s",s);
getch();
}
```

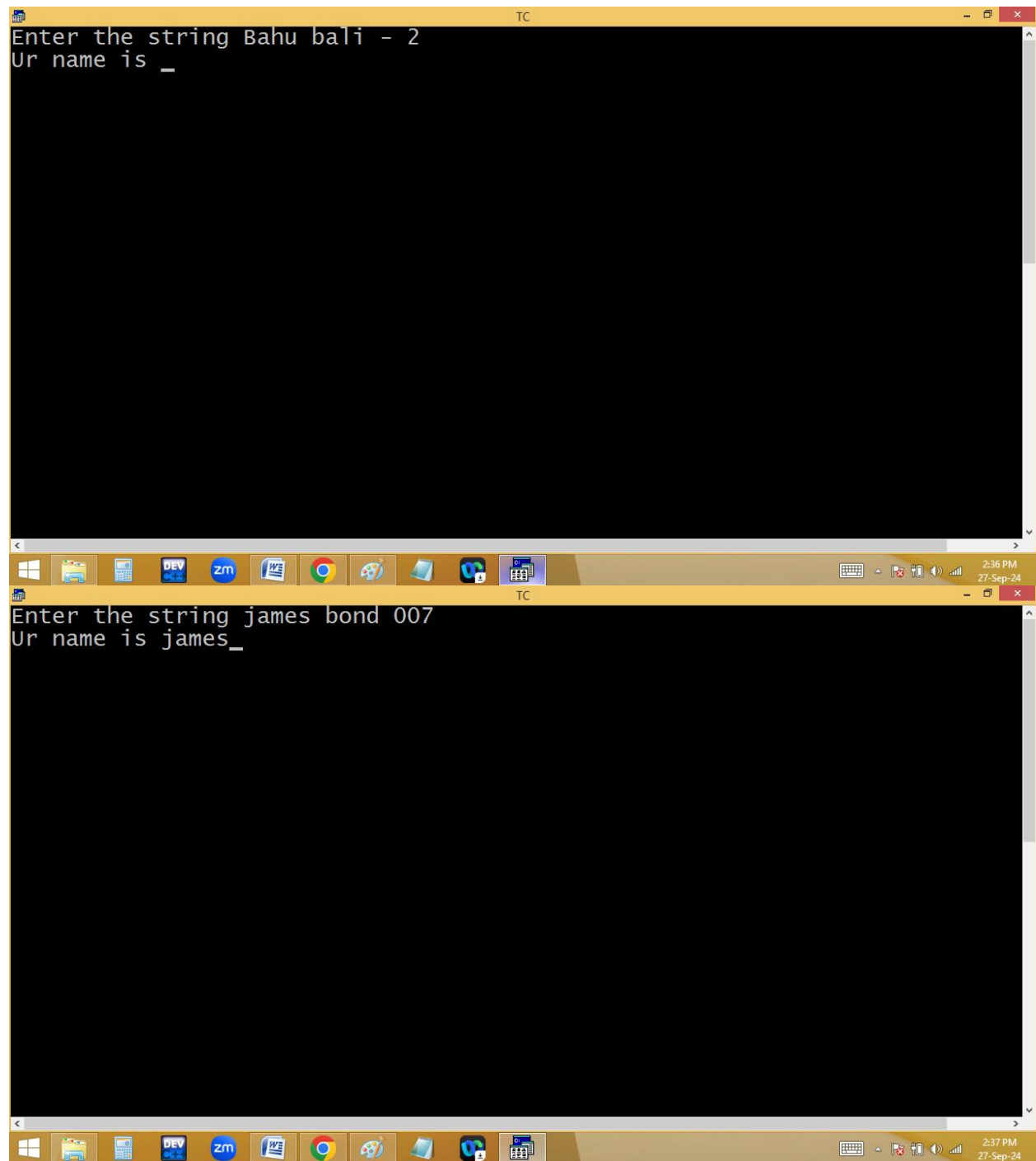
The bottom window shows the execution of the program. It displays the prompt "Enter the string" followed by the user input "james bond 007". The output is "Ur name is james bond _", where the underscore indicates the cursor position after the space. The taskbar at the bottom of both windows shows various application icons and the system clock indicating 2:35 PM on 27-Sep-24.



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 7", "Col 47", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); scanf("%[^0-9 A-Z]",s);
printf("Ur name is %s",s);
getch();
}
```

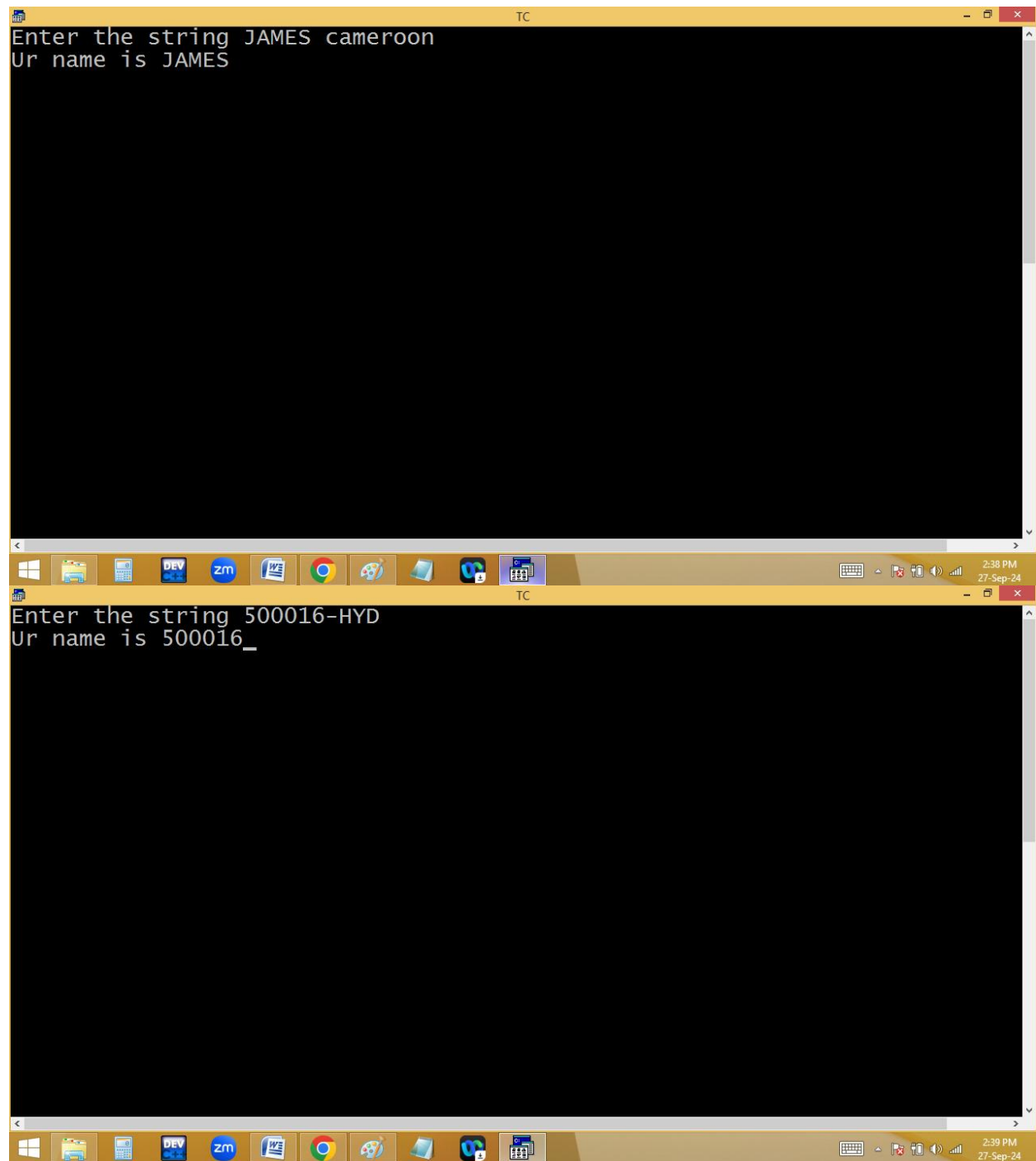
The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Task View, and several application windows including a terminal (DEV), a chat application (zm), a document (WE), Google Chrome, and a game (P). The system tray on the right shows the time as 2:36 PM and the date as 27-Sep-24.

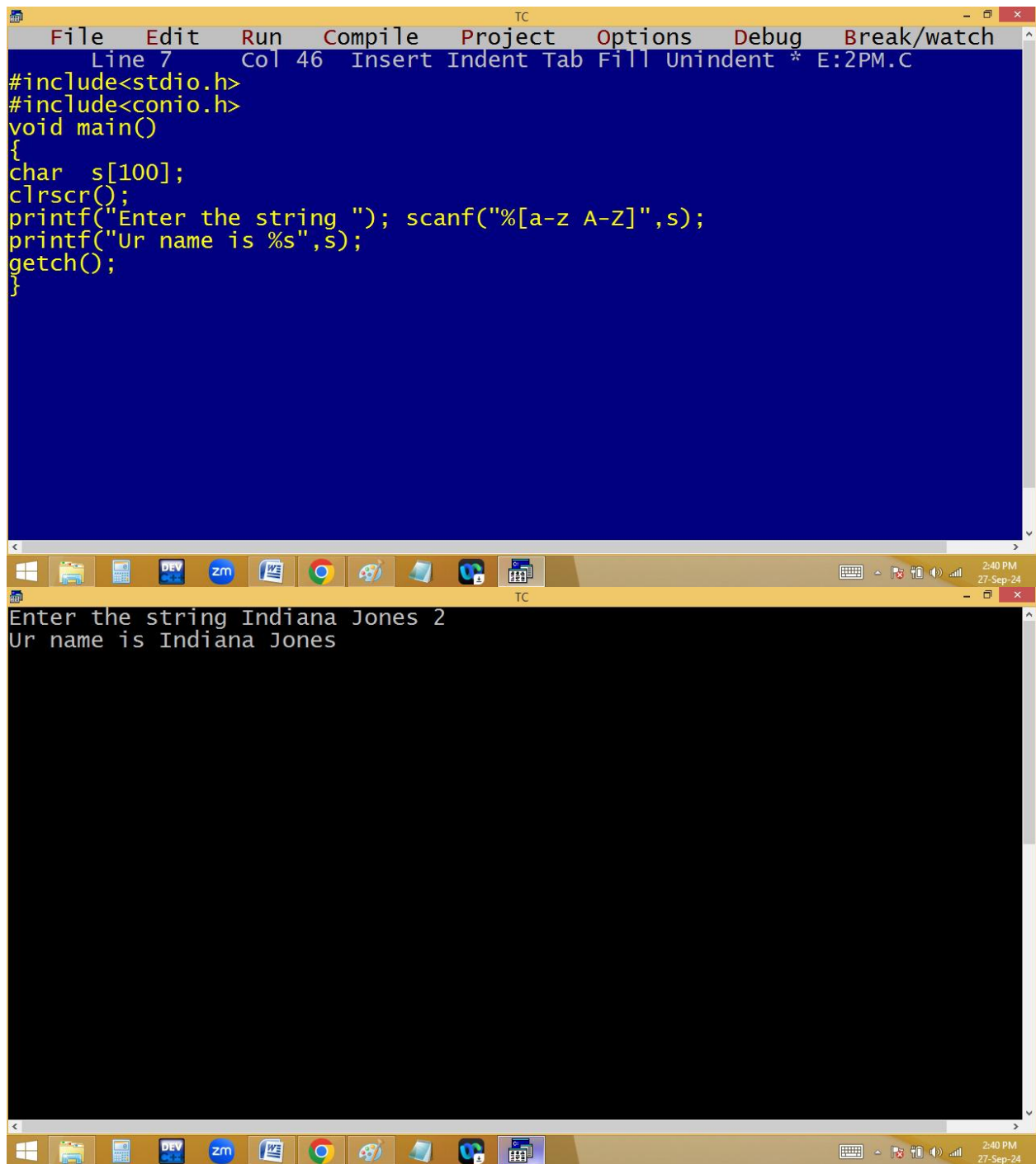


The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that declares a character array `s` of size 100, clears the screen with `clrscr()`, prompts the user to enter a string, reads the input using `scanf`, prints the input using `printf`, and waits for a key press with `getch()`. The bottom screenshot shows the same IDE with the program executed. The output window displays the prompt "Enter the string tomorrow never dies" and the response "Ur name is _". The Windows taskbar at the bottom of both screenshots shows various application icons and the system clock indicating 2:37 PM and 2:38 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 39 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); scanf("%[0-9 A-Z]",s);
printf("Ur name is %s",s);
getch();
}
```

Enter the string tomorrow never dies
Ur name is _





The image shows two windows from the Turbo C++ (TC) IDE. The top window is the code editor, displaying a C program. The code includes `<stdio.h>` and `<conio.h>`, defines a `main` function, declares a character array `s` of size 100, clears the screen with `clrscr()`, prompts the user to enter a string, reads the input using `scanf`, prints the input using `printf`, and waits for a key press with `getch()`. The bottom window is the output console, showing the program's execution. It displays the prompt "Enter the string" followed by the user input "Indiana Jones 2", and then the output "Ur name is Indiana Jones". The Windows taskbar at the bottom shows the time as 2:40 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 46 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); scanf("%[a-z A-Z]",s);
printf("Ur name is %s",s);
getch();
}
```

Enter the string Indiana Jones 2
Ur name is Indiana Jones

gets(): It reads the string with spaces until pressing of enter key.

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that prompts the user for a string, reads it using `gets`, and prints it back using `printf`. The bottom screenshot shows the same IDE with the program's output in a black console window. The user has entered the string "I miss you jaaanu...", and the program has printed "Ur name is I miss you jaaanu...._".

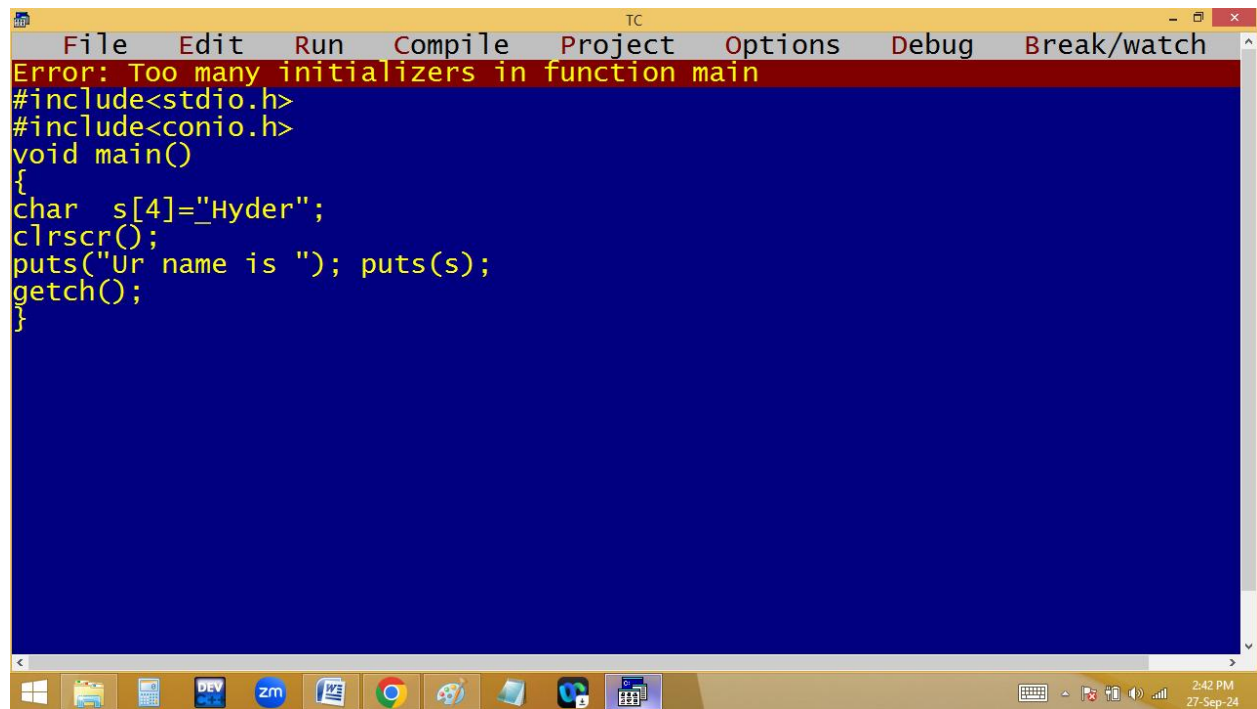
```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
printf("Enter the string "); gets(s);
printf("Ur name is %s",s);
getch();
}
```

Enter the string I miss you jaaanu....
Ur name is I miss you jaaanu...._

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that prompts the user to enter a string, reads it using `gets`, and prints it using `puts`. The bottom screenshot shows the same IDE with the program's output in a black window. The output shows the program's execution flow: it prompts for input, receives the string "I am proud to be indian...", and then prints "Ur name is" followed by the entered string.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100];
clrscr();
puts("Enter the string "); gets(s);
puts("Ur name is "); puts(s);
getch();
}
```

Enter the string
I am proud to be indian...
Ur name is
I am proud to be indian...



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Too many initializers in function main". Below the banner, the following C code is visible in a blue editor area:

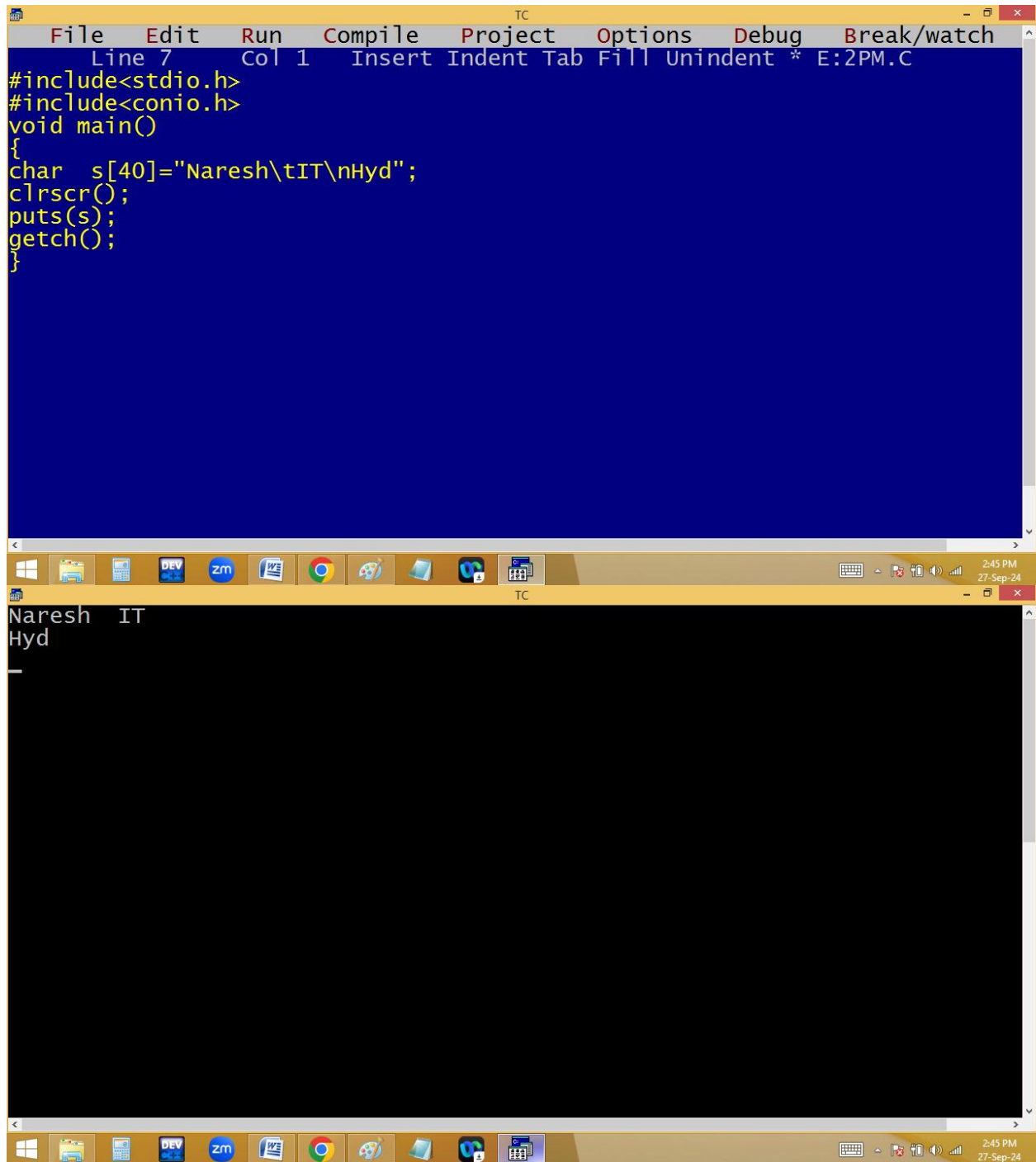
```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[4]="_Hyder";
clrscr();
puts("Ur name is "); puts(s);
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV C++, Zoom, Word, Chrome, Paint, and other applications. The system clock in the bottom right corner shows "2:42 PM" and "27-Sep-24".

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function that declares a character array `s` of size 4, clears the screen, prompts the user for their name, reads the input using `gets`, prints it using `puts`, and waits for a key press using `getch`. The bottom screenshot shows the same IDE with the program executed. The output window displays the prompts and the user's input, "Hyder".

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[4];
clrscr();
printf("Enter ur name "); gets(s);_
puts("Ur name is "); puts(s);
getch();
}
```

Enter ur name Hyder
Ur name is
Hyder



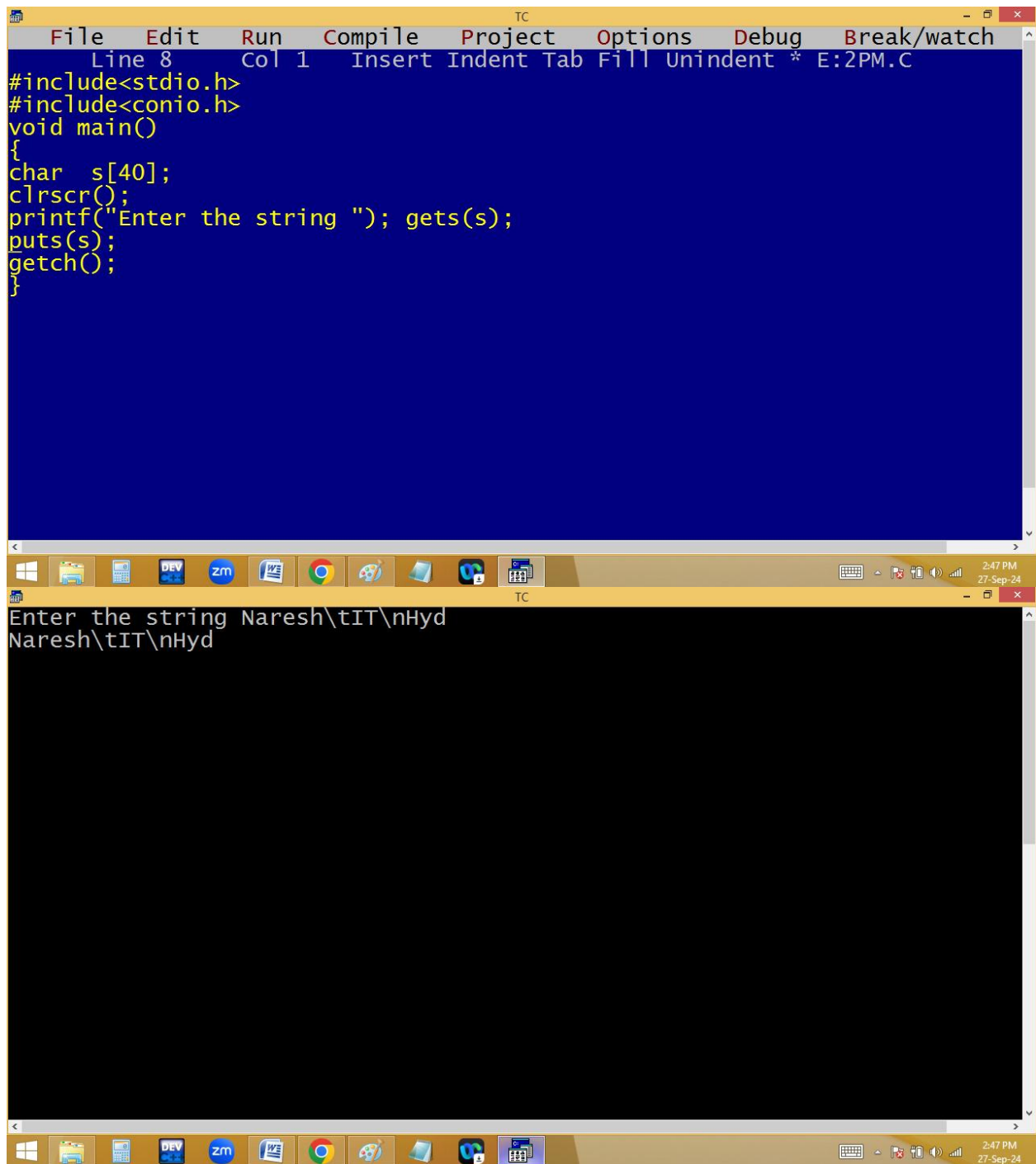
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[40]="Naresh\tIT\nHyd";
clrscr();
puts(s);
getch();
}
```

The bottom window shows the output of the program, which is:

```
Naresh IT
Hyd
_
```

The output is displayed on a black background with white text. The first line is "Naresh IT" and the second line is "Hyd". A cursor is visible on the third line.



The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code includes `<stdio.h>` and `<conio.h>`, defines a `main` function, declares a character array `s` of size 40, clears the screen with `clrscr()`, prints a prompt, reads a string using `gets(s)`, displays it with `puts(s)`, and waits for a key press with `getch()`. The bottom screenshot shows the same IDE with the program's output in a black window. The prompt "Enter the string" is followed by the input "Naresh\tIT\nHyd", which is then printed to the screen. The taskbar at the bottom of both windows shows various application icons and the system clock indicating 2:47 PM on 27-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[40];
clrscr();
printf("Enter the string "); gets(s);
puts(s);
getch();
}
```

Enter the string Naresh\tIT\nHyd
Naresh\tIT\nHyd

Reading multiple string at runtime:

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside `main`, a character array `s[5][40]` is declared, and `clrscr()` is called. The program prompts the user to "Enter 5 names" and uses a `for` loop to read five names into the array `s` using `gets(s[i])`. After the loop, it prints "NAMES" followed by a separator line of dashes, and then prints each name from the array `s` using `puts(s[i])` in a `for` loop. The bottom window shows the program's execution. It displays the prompt "Enter 5 names" and the five names entered: "Virat Kohli", "Dhoni", "Rohith Sharma", "Ravindra jadeja", and "Surya kumar yadav". Below these, it prints "NAMES" followed by a separator line and then the same five names, demonstrating the program's functionality.

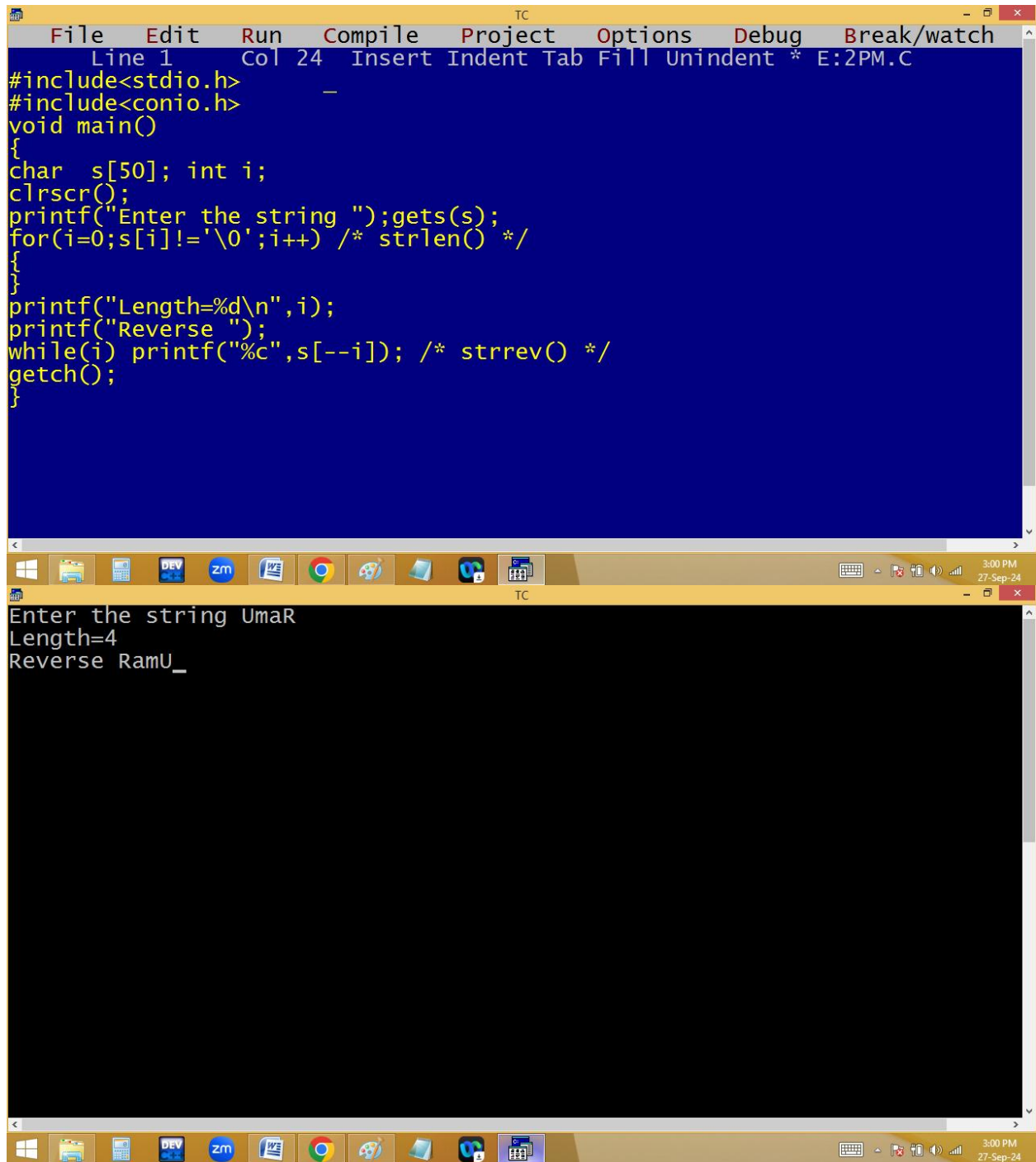
```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[5][40]; int i;
clrscr();
puts("Enter 5 names ");
for(i=0;i<5;i++)gets(s[i]);

puts("NAMES");
puts("-----");
for(i=0;i<5;i++)puts(s[i]);
getch();
}
```

Enter 5 names
Virat Kohli
Dhoni
Rohith Sharma
Ravindra jadeja
Surya kumar yadav
NAMES

Virat Kohli
Dhoni
Rohith Sharma
Ravindra jadeja
Surya kumar yadav

Finding string length and reverse string using user defined functions:



```
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[50]; int i;
clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++) /* strlen() */
{
}
printf("Length=%d\n",i);
printf("Reverse ");
while(i) printf("%c",s[--i]); /* strrev() */
getch();
}
```

Enter the string UmaR
Length=4
Reverse RaMu_

```
TC
Enter the string NaYaN TaRa
Length=10
Reverse aRaT NaYaN
```

```
for( i=0; s[i]!='\0';i++)
{
    strlen()
}
p("Leng=%d\n",i);
    0 ✓
    1
    2
    3
while( i ) p( s[--i] ); strrev()
```

s	A	m	a	R	\0	
	0	1	2	3	4	

--i
3 2 1 0
R a m A

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a program that reverses a string. The code includes `<stdio.h>` and `<conio.h>`, and uses `gets()` to read input, `strlen()` to find the length, and `strrev()` to reverse the string. The bottom screenshot shows the program's execution. It prompts the user to enter the string "nitin level", prints the length as 12, and then prints the reversed string "level nitin_".

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[50]; int i;
clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i];i++); /* strlen() */

printf("Length=%d\n",i);
printf("Reverse ");
while(i) putchar(s[--i]); /* strrev() */
getch();
}
```

Enter the string nitin level
Length=12
Reverse level nitin_

Lower to upper / upper to lower:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[50]; int i;
clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++)
{
if(s[i]>='a' && s[i]<='z') s[i]-=32; /*strupr() */
else if(s[i]>='A' && s[i]<='Z')s[i]+=32; /*strlwr() */
}
puts(s);
getch();
}

Enter the string James Bond - 007
jAMES bOND - 007

TC
3:12 PM
27-Sep-24
```

α B C d
 $65+32$ $98-32$ $99-32$ $68+32$

s	A	b	c	D	\0	
	0	1	2	3	4	

```

for( i=0; s[i]!='\0'; i++)
{
    if(s[i]>='a' && s[i]<='z') s[i]-=32; strupr()
    else if(s[i]>='A' && s[i]<='Z') s[i]+=32; strlwr()
}
p(s);

```

97 122
 65 90
 $0 - A - 65 + 32 = 97 = \alpha$
 $1 - b - 98 - 32 = 66 = B$

Read a string and find out no of vowels, consonants, digits, spaces and special characters.

